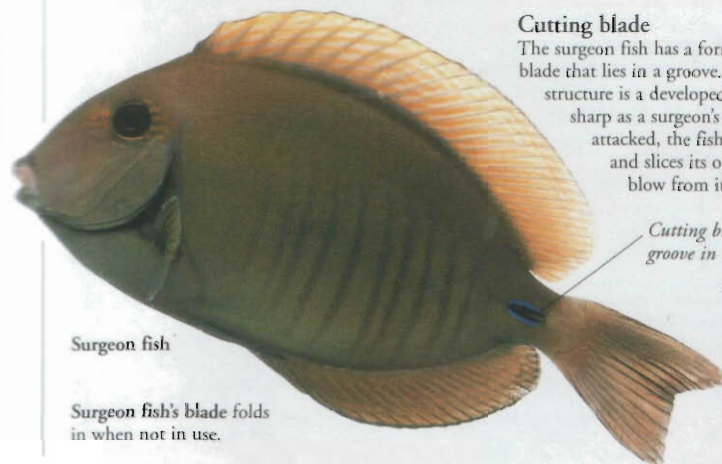


Protection

As well as scales, fish use colour, camouflage, or poisonous spines to protect themselves. The spiny puffer fish can take in water or air and swells up to more than twice its size. Some eels use an electric discharge, while other fish live in shoals, making it difficult for a predator to pick out any one individual.



Surgeon fish

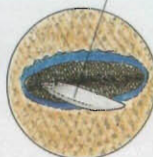
Surgeon fish's blade folds in when not in use.

Cutting blade

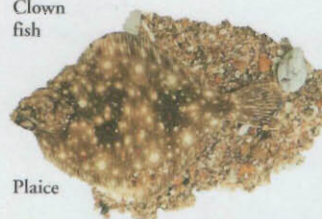
The surgeon fish has a formidable cutting blade that lies in a groove. This defensive structure is a developed scale and is as sharp as a surgeon's scalpel. If attacked, the fish erects the blade and slices its opponent with a blow from its tail.

Cutting blade in a groove in the body

Blade extended



Clown fish



Plaice



Weever fish

Colour

Fish use colour to warn other fish that they are poisonous. Colour also helps some fish hide from predators. The fish's colour depends on its lifestyle. Cave fish have no colour; deep-sea fish are black; open-sea fish are a silvery colour.

Camouflage

Some fish look like the plants among which they live. The triple tail looks like a mangrove leaf, and the leafy sea-dragon and sargassum fish look like seaweed. The plaice can change colour to match the surrounding seabed.

Poison

Several fish are poisonous. The weever fishes are extremely dangerous. They lie partly buried and camouflaged in sand, waiting for food. The spines on their gill covers and dorsal fin can inject poison into anyone who steps on one.



Carnivorous red-bellied piranhas



Piranhas find their food mainly by detecting vibrations in the water.

Food

Many larger fish, including sharks, groupers, pikes, and barracudas, are predators and catch and eat their prey. Most smaller shoal fish feed on plankton, which floats around in the water. Some fish are bottom feeders, such as plaice, while others are grazers, such as the parrot fish, which rasps organisms from rocks.

Piranhas

These South American fish have strong jaws and sharp triangular teeth. A shoal of carnivorous piranhas can strip an animal to the bone in minutes. They are one of the most feared types of fish, but most eat mainly fruit and nuts.

Cleaner fish

Fish called cleaners, such as some wrasses, have "cleaning" stations where they wait for customers. The customers allow the cleaners to remove bits of food and parasites from their skin, gills, fins, and even right inside their mouths.



Cleaner wrasse at work

Angler fish with lure



Angler fish

Angler fish usually live in deep water. They have a dorsal fin ray modified into a fishing line, with a lure on the end to attract their prey. They can swallow fish much larger than themselves.

Reproduction

Most bony fish lay eggs in the water, and these are then fertilized by a male's sperm. The parents usually leave the eggs to their fate, but some species protect their young in their mouths, in pouches, or in nests. Some sharks, such as the dogfish, lay eggs in an egg case, while others bear live young.



Seahorses

Within 10 seconds of mating, the female seahorse transfers 200 fertilized eggs into the brood-pouch of the male. The male carries them around with him for four weeks until they are ready to hatch. The male then "gives birth", and the young seahorses swim away.



Sticklebacks

Male sticklebacks make a nest and attract females by doing a zigzag dance. The female lays her eggs in the nest, and the male fertilizes them. He then protects the nest from any intruder until the young hatch and are ready to fend for themselves.

Male three-spined stickleback

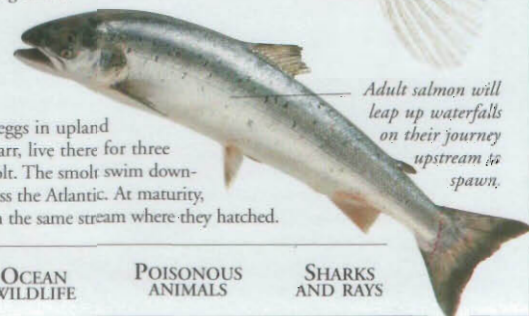


Stickleback eggs

Male stickleback's underside turns red in the breeding season.

Salmon

The Atlantic salmon lays its eggs in upland streams. The young, called parr, live there for three years; then they are called smolt. The smolt swim downriver to the sea and travel across the Atlantic. At maturity, the salmon return to spawn in the same stream where they hatched.



Adult salmon will leap up waterfalls on their journey upstream to spawn.

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WILDLIFE

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AND RAYS